Geometry
Unit 2
Lesson 7 Homework

Name Answer Key_____________
Date______________________
Period_____________________

Complete the proof.

Given: $AB = CD$
Prove: $AC = BD$

	Statement	Reason
1.	$AB = CD$	Given
2.	$BC = BC$	Reflexive Property of Equality
3.	$AB + BC = BC + CD$	Addition Property of Equality
4.	$AB + BC = AC$; $BC + CD = BD$	Segment Addition Postulate
5.	$AC = BD$	Substitution Property of Equality

Complete the paragraph proof.

Given: $\angle 1$ and $\angle 2$ are complementary and $\angle 2$ and $\angle 3$ are complementary.
Prove: $\angle 1 \cong \angle 3$

It is 6. ___given___ that $\angle 1$ and $\angle 2$ are complementary and $\angle 2$ and $\angle 3$ are complementary. By the definition of 7. ___complementary angles___, $m\angle 1 + m\angle 2 = 90°$ and $m\angle 2 + m\angle 3 = 90°$. By the 8. ___transitive___ property, $m\angle 1 + m\angle 2 = m\angle 2 + m\angle 3$.

Then, $m\angle 1 =$ 9. ___$m\angle 3$___ by the subtraction property. Therefore, $\angle 1 \cong \angle 3$ by the 10. ___definition of congruence___.